

A Novel Sidelobe Suppression Method for Multi-Target Sparse Random Array Beamforming

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Abstract—Sparse random arrays reduce hardware costs and naturally eliminate grating lobes, yet their high random sidelobes cause severe target masking in multi-target environments. To mitigate such sidelobe interference without sacrificing angular resolution, this paper proposes a novel sidelobe suppression method based on L_1 norm minimization. In this method, multi-target beamforming is reformulated as a sparse signal recovery problem under an overcomplete dictionary. The proposed method solves the Least Absolute Shrinkage and Selection Operator (LASSO) model through the Iterative Shrinkage-Thresholding Algorithm (ISTA), which can effectively decouple strong and weak targets. Simulation results demonstrate that the proposed scheme allows the sparse random array to achieve sidelobe levels comparable to a uniform linear array (ULA), dropping below -100dB, while simultaneously offering superior multi-target resolution. Furthermore, its low computational complexity and rapid convergence make the approach highly viable for practical engineering applications.

Keywords—Sparse random array, multi-target beamforming, compressed sensing, Iterative Shrinkage-Thresholding Algorithm

I. INTRODUCTION

Sparse arrays can achieve the angular resolution of a full array with fewer elements and are widely applied in radar [1]-[3], wireless communications [4]-[6], and sonar [7][8]. Sparse random arrays are a common type of sparse array [4][9]. By breaking the uniformity of element positions, they overcome the grating lobe problem inherent to deterministic sparse arrays [6][10]-[13]. However, high peak sidelobe levels (PSL) are an inevitable issue in sparse random arrays [14][15]. In multi-target beamforming, high sidelobes tend to mask weak targets and degrade the angular resolution capability of the system. Therefore, it is essential to investigate sidelobe suppression methods for multi-target beamforming in sparse random arrays.

Traditional multi-target beamforming methods are mainly based on the Linear Constrained Minimum Variance (LCMV) criterion, which realizes multi-mainlobe control by imposing multiple point constraints [16]-[18]. However, the LCMV algorithm is generally designed for full arrays and thus not suitable for sparse arrays. For non-uniform arrays such as sparse random arrays, sidelobe suppression is essential to guarantee reliable multi-target detection [17][18]. Existing sidelobe suppression approaches primarily rely on weighted convex optimization, which suppresses the global sidelobe level by adjusting the amplitude weights of array elements. Nevertheless,

under sparse array configurations, such methods usually sacrifice mainlobe width, namely angular resolution [19][20]. To strike a balance between angular resolution and sidelobe level, Compressed Sensing (CS) theory has been introduced into beamforming research [21]-[24]. Within the CS methods, sparse recovery algorithms can be grouped into three types: greedy algorithms, Sparse Bayesian Learning (SBL), and convex optimization. Greedy methods such as Orthogonal Matching Pursuit (OMP) are computationally light but highly noise-sensitive, with performance degrading markedly at low SNR. SBL adapts model complexity and provides good robustness, yet relies on expensive matrix inversions. Convex optimization delivers high reconstruction accuracy, though classical solvers can incur prohibitive computational costs for large dictionaries. Therefore, the mainstream research shifts to constructing an overcomplete angular dictionary. By virtue of L_1 norm minimization, the beamforming problem is transformed into the Least Absolute Shrinkage and Selection Operator (LASSO) model for solution. As an efficient tool for solving the LASSO model, the Iterative Shrinkage-Thresholding Algorithm (ISTA) combine high computational efficiency with superior nonlinear denoising capability, and have become a core technique for ultra-high-resolution signal reconstruction [27][28]. Nevertheless, although relevant algorithms have been applied to specific array configurations [29][30], research targeting general sparse random arrays remains insufficient for challenging scenarios such as low signal-to-noise ratio and multi-target conditions.

This paper proposes an efficient sidelobe suppression method for multi-target beamforming in sparse random arrays by developing a lightweight, inversion-free CS- L_1 -ISTA algorithm based on sparse recovery theory. Specifically, this method overcomes the noise sensitivity typical of the OMP method and avoids the matrix inversion bottleneck inherent to SBL and LCMV. Benefiting from low computational complexity and fast convergence, the proposed scheme is highly suitable for multi-target resolution scenarios in complex environments.

The paper is organized as follows: Section II describes the mathematical optimization model. Section III details the proposed CS- L_1 -ISTA algorithm. Section IV presents simulations results, and Section V summarize several conclusions.

II. MATHEMATICAL OPTIMIZATION MODEL

Assume there are M array elements and K far-field narrowband signal sources in space. The signal received by the m -th antenna element at time t is expressed as:

$$y_m(t) = \sum_{k=1}^K s_k(t) e^{-j2\pi \frac{d_m \sin(\theta_k)}{\lambda}} + n_m(t) \quad (1)$$

where $s_k(t)$ is the complex envelope of the k -th signal, θ_k is the Direction of Arrival (DOA) of the k -th signal, d_m is the position of the m -th array element relative to the reference point, λ is the wavelength, and $n_m(t)$ is the additive noise. Stack the received signals of all array elements into a vector:

$$\mathbf{y}(t) = \mathbf{A}(\theta) \mathbf{s}(t) + \mathbf{n}(t) \quad (2)$$

where $\mathbf{y}(t)$ is the $M \times 1$ observation vector, $\mathbf{s}(t)$ is the $K \times 1$ signal vector, $\mathbf{n}(t)$ is the $M \times 1$ noise vector, $\mathbf{A}(\theta)$ is the $M \times K$ steering vector matrix. The steering vector is expressed as:

$$\mathbf{a}(\theta_k) = \left[1, e^{-j2\pi \frac{d_2 \sin(\theta_k)}{\lambda}}, \dots, e^{-j2\pi \frac{d_M \sin(\theta_k)}{\lambda}} \right]^T \quad (3)$$

In practical, these directions are unknown. The core idea of sparse recovery is to discretize the entire angular space into N grid points, and assume that signals can only come from these grid directions.

$$\Theta = \{\theta_1, \theta_2, \dots, \theta_N\} \quad (4)$$

Now, an over-complete steering vector matrix Φ is constructed, which includes all possible grid directions:

$$\Phi = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_N)] \quad (5)$$

Φ is an $M \times N$ matrix, where $M < N$, representing an underdetermined system. Define the $N \times 1$ sparse vector:

$$\mathbf{x} = [x_1, x_2, \dots, x_N]^T \quad (6)$$

As there are only K real signals, the sparse vector contains only K non-zero elements, while the remaining $N-K$ array elements are zero. The observation model is thus rewritten as:

$$\mathbf{y} = \Phi \mathbf{x} + \mathbf{n} \quad (7)$$

From compressed sensing theory, it is known that the most direct method to recover the sparse signal is to solve the following optimization problem:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_0 \quad \text{subject to} \quad \|\mathbf{y} - \Phi \mathbf{x}\|_2 \leq \varepsilon \quad (8)$$

where $\|\mathbf{x}\|_0$ denotes the L_0 norm, which represents the number of non-zero array elements in $\mathbf{x}$, ε is the noise tolerance. However, L_0 norm minimization is an NP-hard problem. Its computational complexity grows exponentially with N , making it unsolvable in practice. The core contribution of compressed sensing is that, under certain conditions, L_1 norm minimization is equivalent to L_0 norm minimization. Thus, the problem is relaxed to:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_1 \quad \text{subject to} \quad \|\mathbf{y} - \Phi \mathbf{x}\|_2 \leq \varepsilon \quad (9)$$

where $\|\mathbf{x}\|_1 = \sum_{i=1}^N |x_i|$ denotes the L_1 norm. Geometrically,

unlike the smooth L_2 norm ball which yields dense solutions, the L_1 norm ball possesses sharp corners situated on the coordinate axes. The constraint hyperplane is highly likely to intersect at these corners, inherently promoting sparse solutions by forcing unselected elements to zero.

III. THE PROPOSED CS-L1-ISTA ALGORITHM

To tackle the severe target masking problem induced by high random sidelobes, this paper reformulates multi-target beamforming as a sparse spatial spectrum recovery problem. Since real targets are sparsely distributed in the spatial domain, such a characteristic naturally fits the framework of CS. The LASSO model provides a unified formulation. It converts the process of separating discrete targets from dense interference backgrounds into a tractable optimization problem, enabling simultaneous target reconstruction and sidelobe suppression without sacrificing angular resolution. Consider the LASSO form:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \frac{1}{2} \|\mathbf{y} - \Phi \mathbf{x}\|_2^2 + \alpha \|\mathbf{x}\|_1 \quad (10)$$

where α is a regularization parameter that balances the data fitting term and the sparsity penalty term. The objective function consists of two parts:

$$g(\mathbf{x}) = \frac{1}{2} \|\mathbf{y} - \Phi \mathbf{x}\|_2^2 \quad (11)$$

$$h(\mathbf{x}) = \alpha \|\mathbf{x}\|_1 \quad (12)$$

Since the objective function is non-differentiable, the proximal gradient descent method is used. Gradient calculation is given as follows:

$$\nabla g(\mathbf{x}) = \Phi^H (\Phi \mathbf{x} - \mathbf{y}) \quad (13)$$

$$\mathbf{x}^{(k+1)} = \text{prox}_{\eta h}(\mathbf{x}^{(k)} - \eta \nabla g(\mathbf{x}^{(k)})) \quad (14)$$

where η is the step size and $\text{prox}(\cdot)$ is the proximal operator.

For the L_1 norm, the proximal operator is given by the soft-thresholding function:

$$\text{prox}_{\eta \lambda \|\cdot\|_1}(z_i^{(k)}) = \text{sign}(z_i^{(k)}) \cdot \max(|z_i^{(k)}| - \eta \lambda, 0) \quad (15)$$

$$\mathbf{z}^{(k)} = \mathbf{x}^{(k)} - \eta \nabla g(\mathbf{x}^{(k)}) \quad (16)$$

The algorithm iteratively alternates between gradient descent and soft thresholding until the convergence condition is satisfied.

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_2 < \epsilon \quad (17)$$

After obtaining the optimal sparse solution $\hat{\mathbf{x}}$ via the above CS-L1-ISTA algorithm, the final multibeam spectrum reconstruction can be reconstructed by taking the squared magnitude of each element of $\hat{\mathbf{x}}$

$$P(\theta_i) = |\hat{x}_i|^2, \quad i = 1, \dots, N \quad (18)$$

The sparse recovery method does not rely on sample covariance matrix estimation. It directly processes a single snapshot or a small number of snapshots. If there are multiple solutions that can fit the data $\mathbf{y}$ well, L_1 norm minimization will

select the sparsest one. The solutions corresponding to high sidelobes contain multiple non-zero elements, which do not conform to the sparsity prior and are therefore not selected. For reproducibility, Algorithm 1 summarizes the proposed CS-L1-ISTA method. It enforces sparsity by iteratively applying gradient descent and soft-thresholding to the input data and dictionary.

Algorithm 1 : CS-L1-ISTA

Input: $y, \Phi, K_{\max}, \epsilon$.

Output: $P(\theta)$.

Initialization: Set $\mathbf{x}^{(0)} = \mathbf{0}$, and select the step size

$$\eta < \frac{2}{\|\Phi^H \Phi\|_2}.$$

For $k = 0 : K_{\max}$ **do**

$$\mathbf{z}^k = \mathbf{x}^k - \eta \Phi^H \Phi \mathbf{x}^k - y$$

For $i = 1 : N$ **do**

$$x_i^{k+1} = \text{sign } z_i^k \cdot \max |z_i^k| - \eta \alpha, 0$$

End For

If $\|\mathbf{x}^{k+1} - \mathbf{x}^k\|_2 < \epsilon$ **then**

Break loop

End If

End For

After obtaining the sparse solution $\hat{\mathbf{x}}$, construct $P(\theta)$ as follows:

$$P(\theta_i) = |\hat{x}_i|^2, \quad i = 1, \dots, N$$

IV. SIMULATION RESULTS

To validate the effectiveness of the proposed CS-L1-ISTA method for sparse random array beamforming, a comprehensive simulation experiment is constructed in this section. Detailed analysis and discussion of the parameter settings, experimental results, and performance are provided. The core parameters for the simulation experiment are listed in TABLE I, which together define a typical millimeter-wave radar detection scenario. Three independent sources are simulated at DOAs of -30° , 10° , and 45° . The SNR is set to 20dB. For the L_1 minimization framework, the regularization parameter α is set to 0.1~1, the maximum number of iterations is limited to 100, and the convergence tolerance ϵ is 1×10^{-6} .

TABLE I. SIMULATION PARAMETER SETTINGS

Parameter Name	Parameter Value
Carrier Frequency (f_c)	10 GHz
Speed of Light (c)	3×10^8 m/s
Wavelength (λ)	0.03 m

Fig. 1 presents a comparison of array configurations. The 8-element sparse array with positions [1, 3, 6, 8, 10, 12, 14, 16] is randomly generated to demonstrate the algorithm's robustness on general unoptimized array geometries. This layout maintains an aperture of $15d$, matching the angular resolution of a 16-element ULA. Meanwhile, its non-uniform element spacing

breaks spatial periodicity, which serves as a fundamental physical prerequisite for the CS-L1-ISTA method to effectively suppress high sidelobes.

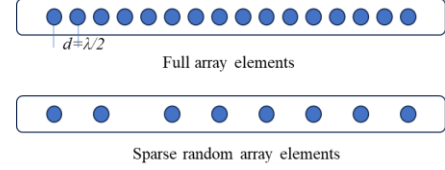


Fig. 1. The array structure.

A. Multi-Target Beamforming Performance Comparison

As presented in Fig. 2, despite having the same physical aperture, the 8-element sparse array achieves a narrower main lobe than the 16-element uniform linear array (ULA). Conventional beamforming is constrained by the Rayleigh limit, while the proposed CS-L1-ISTA method formulates the problem as sparse signal reconstruction. This method breaks through the limitation of physical aperture to achieve super-resolution capability, and can effectively alleviate the inherent main lobe masking effect of multiple targets in conventional approaches.

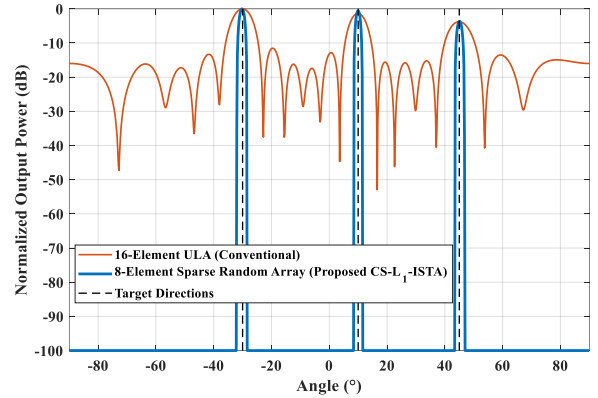


Fig. 2. Multi-Target Beamforming Performance Comparison.

The proposed CS-L1-ISTA method demonstrates a significant advantage in sidelobe suppression. Compared to the continuous, rippling sidelobe leakage across the entire spatial spectrum of the 16-element uniform linear array, which peaks at approximately -12 dB, the proposed 8-element sparse array not only successfully suppresses residual spurious peaks below -100 dB, but also leverages the nonlinear soft-thresholding property of the LASSO model to completely clear background interference in the vast majority of non-target regions down to the -100 dB noise floor limit. This exceptional spatial background cleansing capability fundamentally eliminates the weak target masking effect in multi-target environments.

B. Parameter Sensitivity Analysis

Fig. 3 further investigates the impact of key parameters on algorithm performance.

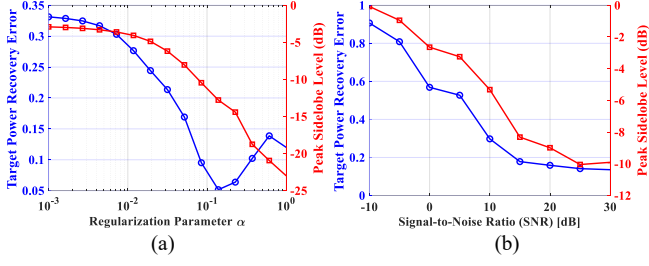


Fig. 3. (a) The impact of α on algorithm performance. (b) The impact of SNR on algorithm performance.

As illustrated in Fig. 3, the proposed CS-L1-ISTA method demonstrates exceptional robustness and optimal parameter trade-offs. Fig. 3(a) reveals a critical V-shaped trend in target power recovery error caused by the inherent shrinkage bias of L_1 minimization. It identifies $\alpha=0.1\sim 1$ as the optimal operating range, perfectly balancing deep sidelobe suppression with accurate mainlobe amplitude preservation. Furthermore, Fig. 3(b) displays a clear transition into an error floor. The algorithm converges rapidly in noise-limited environments at low to medium SNRs, whereas its accuracy at higher SNRs becomes constrained by the physical manifold limits of the sparse array rather than thermal noise. This behavior effectively verifies the method's capability to reliably decouple multiple targets in complex scenarios.

C. Algorithm Convergence and Performance Metrics

As depicted in Fig. 4, the CS-L1-ISTA algorithm reduces the residual to the order of 10^{-2} within about 10 iterations and stabilizes within 20 iterations, demonstrating its computational efficiency and rapid convergence, making it suitable for practical engineering applications.

As shown in Fig. 5, the proposed CS-L1-ISTA method was benchmarked against the Bartlett with a rectangular window, LCMV based on the sample covariance matrix with 0.001 diagonal loading, and Robust SBL. The evaluation used 100 snapshots on both the proposed array geometry and an 8-element nested array. For a setup with M elements, N grid points, L snapshots, and I iterations, the computational costs of Bartlett, LCMV, and the proposed CS-L1-ISTA method scale as $O(NM)$, $O(LM^2 + M^3)$, and $O(IMN)$ operations, respectively. By sidestepping the costly matrix inversion in LCMV, CS-L1-ISTA achieves linear scaling and is readily parallelizable. More fundamentally, Fig. 5 illustrates a conceptual shift from merely suppressing sidelobes to circumventing the ambiguity entirely. CS-L1-ISTA directly reconstructs a sidelobe-free spatial spectrum and strikes a favorable computational balance: it avoids the prohibitive complexity of SBL while delivering better performance than LCMV in real-time scenarios.

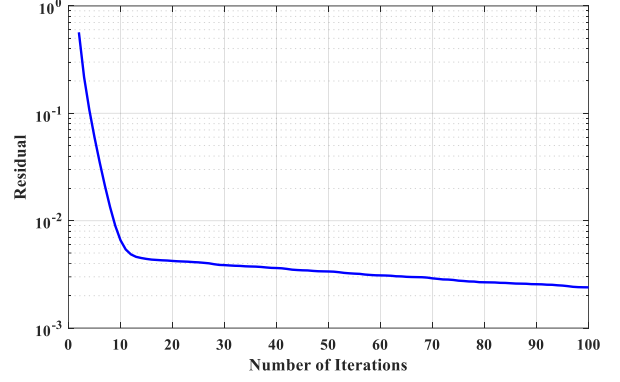


Fig. 4. Convergence Curve of the CS-L1-ISTA Algorithm.

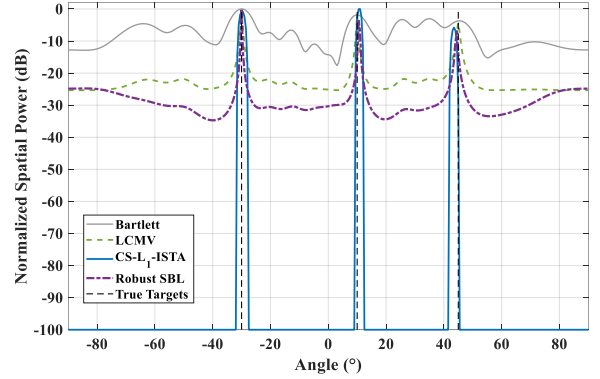


Fig. 5. Performance Reference with different methods.

V. CONCLUSION

This paper formulates multi-target beamforming for sparse random arrays as a compressed sensing (CS) reconstruction problem. To tackle the limitation of multi-target resolution degradation induced by high random sidelobes, a lightweight CS-L1-ISTA sidelobe suppression method is proposed. The proposed method avoids high-complexity matrix inversion and achieves accurate decoupling of strong and weak targets, with superior performance and robustness verified by numerical simulations. ISTA serves as a reliable and low-computation baseline solver, whereas its slow convergence requires the integration of accelerated algorithms such as Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) or Alternating Direction Method of Multipliers (ADMM) to meet the strict real-time requirement of practical radar systems. Furthermore, to cope with the inherent off-grid basis mismatch, future research will introduce continuous-domain sparse recovery techniques, including atomic norm minimization and dynamic grid refinement. These schemes can effectively eliminate energy leakage and improve estimation precision under practical conditions such as mutual coupling and phase errors.

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